

21 Batch reactor energy balance

Let's derive the energy balance for the batch reactor. Starting from the general energy balance equation:

$$\frac{d\hat{E}_{sys}}{dt} = \dot{Q} + \dot{W}_s + \sum_i \dot{F}_i \hat{H}_i|_{in} - \sum_i \dot{F}_i \hat{H}_i|_{out}$$

Where $\hat{E}_{sys}$ is the total energy of all components in the system

$$\hat{E}_{sys} = \sum N_i \hat{E}_i = \sum N_i \hat{U}_i + \sum N_i \hat{K} \hat{E}_i + \sum N_i \hat{P} \hat{E}_i$$

$\nwarrow$ internal $\nwarrow$ kinetic $\nwarrow$ potential

Simplifying, $\hat{E}_{sys} = \sum N_i \hat{U}_i$ since we ultimately seek $\Delta \hat{E}_{sys} \approx \Delta \hat{U}$

Since we ultimately seek $\Delta \hat{E}_{sys}$, $\Delta \hat{P} \hat{V}$ is also $\ll \Delta \hat{U}$ & hence PV can be neglected.

Applying the product rule $\frac{d\hat{E}_{sys}}{dt} = \left[\sum N_i \frac{d\hat{U}_i}{dt} + \sum \hat{U}_i \frac{dN_i}{dt} \right]_{sys}$

$$\text{So, } \hat{E}_{sys} = \sum N_i \hat{H}_i$$

$$\therefore \frac{d\hat{E}_{sys}}{dt} = \frac{d}{dt} \sum N_i \hat{H}_i$$

Recalling that $H_i = H_i^0(T_{ref}) + \int_{T_{ref}}^T C_{p,i} dT$

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$$\frac{dH_i}{dt} = \frac{d}{dt} \left(H_i^0(T_{ref}) + \int_{T_{ref}}^T C_{p,i} dT \right) = \frac{d}{dt} \int_{T_{ref}}^T C_{p,i} dT = C_{p,i} \frac{dT}{dt} \quad (\text{assuming } C_{p,i} \text{ are constant})$$

$$\text{Then: } \dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{p,i} \frac{dT}{dt} + \sum H_i \frac{dN_i}{dt}$$

Recall $\frac{dN_i}{dt}$ is the mole balance and equals $r_i V$

$$\text{Then } \dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{p,i} \frac{dT}{dt} + \sum H_i r_i V. \quad r_i = \frac{r_i}{r_A} \quad r_A = \frac{r_i}{|r_A|} (-r_A)$$

$$\sum N_i C_{p,i} \frac{dT}{dt} + \sum H_i \frac{r_i}{|r_A|} (-r_A V)$$

Recalling that $\sum v_i H_i = \Delta H^{rxn}$ per mole A the energy balance becomes

$$\dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = \sum N_i C_{pi} \frac{dT}{dt} + \Delta H^{rxn} (-r_A V)$$

Solving for $\frac{dT}{dt}$:

$$\frac{dT}{dt} = \frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{\sum N_i C_{pi}}$$

Where $(\Delta H^{rxn})(r_A V)$ is heat generated $= \dot{Q}_{gb}$

And $\dot{Q}$ is heat removed $= \dot{Q}_{rb}$

$$\dot{Q} = UA(T_a - T)$$

Batch reactor energy balance written in terms of X:

$$\text{Recall } N_i = N_{A0} \left(\theta_i - \frac{v_i}{v_A} X \right)$$

$$\text{Then, } \frac{dT}{dT} = \frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{N_{A0} \left(\sum C_{pi} \left(\theta_i - \frac{v_i}{v_A} X \right) \right)}$$

$$\text{Recall } \sum \frac{v_i}{|v_A|} C_{Pi} = \Delta C_p \text{ per mole A}$$

$$\text{Then, } \frac{dT}{dT} = \frac{\dot{Q} + \dot{W}_s + (\Delta H^{rxn})(r_A V)}{N_{A0} \left(\sum \theta_i C_{pi} + \Delta C_p X \right)}$$

where often we neglect $\dot{W}_s$

Let's do some examples:

- The reaction $A + B \rightarrow C$ is carried out in a batch reactor. $C_{PA} = 35 \text{ cal/mol/K}$, $C_{PB} = 18 \text{ cal/mol/K}$, $C_{PC} = 46 \text{ cal/mol/K}$. Determine ΔC_p .

$$\begin{aligned}\Delta C_p &= \sum \frac{\nu_i}{|\nu_A|} C_{pi} = -\frac{1}{1} C_{pA} + \frac{-1}{1} C_{pB} + \frac{1}{1} C_{pC} = \\ &= C_{pC} - C_{pA} - C_{pB} = 46 \frac{\text{cal}}{\text{mol K}} - 35 \frac{\text{cal}}{\text{mol K}} - 18 \frac{\text{cal}}{\text{mol K}} \\ &= -7 \frac{\text{cal}}{\text{mol K}}\end{aligned}$$

- For the same reaction, the inlet molar flowrates are 1 mol A/min, 2 mol B/min, and 0 mol C/min. Determine θ_i .

$$\theta_i = \frac{F_{i0}}{F_{A0}} \quad \text{assuming A is the LR}$$

$$\theta_A = 1, \quad \theta_B = 2, \quad \theta_C = 0$$

- The reaction is carried out adiabatically. What is the energy balance?

Two options

$$\textcircled{1} \quad \frac{dT}{dt} = \frac{(\Delta H_{rxn})(r_A V)}{N_{Ao} (\sum \theta_i C_{pi} + \Delta C_p X)}$$

$$\textcircled{2} \quad T = T_0 + \frac{(-\Delta H_{rxn})X}{\sum \theta_i C_{pi} + \Delta C_p X}$$